HW2

1.2-2 a) 4 temp, 5 pressures, 2 catalysts

$4 \times 5 \times 2 = 40$ experiments

b) each factor is restricted to 2 levels
2^3 design $\rightarrow 2^3 = 8$ experiments

1.2-4 a) Choose flavor (4 ways) x choose 3 toppings from 6

$4 \times \binom{6}{3} = 4 \times 20 = 80$ sundaes

b) $4 \times 2^6 = 4 \times 64 = 256$ sundaes

1.2-5 a) No repeats (4 letter arrangement of 4 distinct letters): $4! = 24$

b) $4^4 = 256$ (4 position = 4 choices)

1.2-9 a) AAAA or NNNN $\rightarrow$ 2 orders

b) $\binom{4}{3} = 4$ orderings, Total $2 \times 4 = 8$

c) $\binom{5}{3} = 10$ Total $= 2 \times 10 = 20$

d) $\binom{6}{3} = 20$ Total $2 \times 20 = 40$

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1.2-16 Total $= \binom{52}{9}$

a) $P = \dfrac{\binom{19}{3}\binom{33}{6}}{\binom{52}{9}} \approx 0.2917 \approx 29.17\%$

b) $P = \dfrac{\binom{19}{3}\binom{10}{2}\binom{7}{1}\binom{5}{1}\binom{6}{2}}{\binom{52}{9}}$

$\approx 0.00622 \approx 0.622\%$

1.3-3 a) $P(A_1 \cap B_1) = \dfrac{\text{count}(A_1 \cap B_1)}{35} = \dfrac{5}{35} = \dfrac{1}{7}$

b) $= P(A_1) + P(B_1) - P(A_1 \cap B_1) = \dfrac{12}{35} + \dfrac{19}{35} - \dfrac{5}{35}$

$= 26/35$

c) $= \dfrac{P(A_1 \cap B_1)}{P(B_1)} = \dfrac{5/35}{19/35} = \dfrac{5}{19}$

d) $= \dfrac{P(A_2 \cap B_2)}{P(A_2)} = \dfrac{9/35}{23/35} = \dfrac{9}{23}$

e) $P(A_2 \mid B_1) = \dfrac{14}{19} \approx 0.7368$

$P(A_2 \mid B_2) = 9/16 = 0.5625$

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Select a student with left thumb on top (B1), that gives a higher chance ($\sim 73.7\%$) of the student being right eye dominant than choosing someone with right thumb on top ($\sim 56.3\%$)

1.3-6 $982 - 334 = 648$

$334 = 221 - 111 = 110$.

$P(\text{died} \mid \text{neither parent had disease}) = \frac{110}{648} = \frac{55}{324}$

≈ 0.1698

1.3-12 a) With 1 blue, 17 red, the probability that the blue chip is 1st, 5th, or last is the same

$P = \frac{1}{18}$ (symmetry, random permutation)

b) $P = \frac{2}{18} = \frac{1}{9}$ every position → any position is equally good

1.3-14 a) $P(5,3) = 5 \cdot 4 \cdot 3 = 60$

b) $5^3 = 125$

1.3-16 Bowl A: 3 red, 2 white (total 5)
Bowl B: initially: 4 red, 3 white (total 7)

Transfer red $P(\text{red from A}) = 3/5$. After transfer B has $4 + 1 = 5$ red, 3 white → total 8, so →

P(red from B | red transferred) = 5/8

Transfer white : P(white from A) = 2/5. After B has 4 red & 3+1 = 4 white → total of 8

P(red from B | white transferred) = $\frac{4}{8} = \frac{1}{2}$

$$\therefore P = \frac{3}{5} \cdot \frac{5}{8} + \frac{2}{5} \cdot \frac{1}{2} = \frac{3}{8} + \frac{1}{5} = 0.575$$

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